SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I P. HARINI (192521402) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Harini

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%	2	2	1	2
Method Selection	20%	2	1	2	2
Solution Process	25%	2	2	2	2
Accuracy of Calculation	20%	2	2	1	2
Interpretation of Results	15%	1	1	2	1

Total Marks 34 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE CODE: CSA0710

COURSE OUTCOME COVERED

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Assessment Tool Weightage: 40%

Code of Conduct:

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Total Marks /40 ***Signature of the Faculty***

Annexure A

ANALYTICAL PROBLEM SOLVING

Q1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

Solution:

Given: Network block = 192.168.20.0/24; Lab A = /26, Lab B = /27, Lab C = /28.

To find: Network address, valid host range, broadcast address and number of usable hosts of each lab.

Formulae used:

(i) Host bits = 32 − prefix length

(ii) Block size = $2^{\text{host bits}}$

(iii) Usable hosts = $2^{\text{host bits}} - 2$

(iv) Broadcast address = network address + block size − 1

(v) Valid host range = (network address + 1) to (broadcast address − 1)

Step 1: Subnet masks

/26 = 11111111.11111111.11111111.11000000 = 255.255.255.192

/27 = 11111111.11111111.11111111.11100000 = 255.255.255.224

/28 = 11111111.11111111.11111111.11110000 = 255.255.255.240

Step 2: Block size and usable hosts

Lab A (/26): Host bits = 32 − 26 = 6; block size = $2^6 = 64$; usable hosts = 64 − 2 = 62

Lab B (/27): Host bits = 32 − 27 = 5; block size = $2^5 = 32$; usable hosts = 32 − 2 = 30

Lab C (/28): Host bits = 32 − 28 = 4; block size = $2^4 = 16$; usable hosts = 16 − 2 = 14

Step 3: Allocation of subnets (largest block first, so that the subnets do not overlap)

Lab A: starts at 192.168.20.0

Network address = 192.168.20.0 /26

Broadcast address = 0 + 64 − 1 = 63 → 192.168.20.63

Valid host range = 192.168.20.1 to 192.168.20.62

Usable hosts = $2^6 - 2 = 62$

Lab B: starts at 0 + 64 = 64

Network address = 192.168.20.64 /27

Broadcast address = 64 + 32 − 1 = 95 → 192.168.20.95

Valid host range = 192.168.20.65 to 192.168.20.94

Usable hosts = $2^5 - 2 = 30$

Lab C: starts at 64 + 32 = 96

Network address = 192.168.20.96 /28

Broadcast address = 96 + 16 − 1 = 111 → 192.168.20.111

Valid host range = 192.168.20.97 to 192.168.20.110

Usable hosts = $2^4 - 2 = 14$

Final Answer:

Lab A: 192.168.20.0/26 | .1 – .62 | Broadcast .63 | 62 hosts

Lab B: 192.168.20.64/27 | .65 – .94 | Broadcast .95 | 30 hosts

Lab C: 192.168.20.96/28 | .97 – .110 | Broadcast .111 | 14 hosts

Conclusion: The three subnets are continuous and do not overlap. Addresses 192.168.20.0 to 192.168.20.111 (112 addresses) are used, and 192.168.20.112 to 192.168.20.255 (144 addresses) remain free for future use.

Q2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Solution:

Given: System 1 = 192.168.1.130; System 2 = 192.168.1.190; Subnet mask = 255.255.255.192.

To find: Network address, broadcast address, subnet range, valid host range, whether each IP lies in its subnet, and usable hosts.

Step 1: Prefix length and block size

255.255.255.192 = 11111111.11111111.11111111.11000000

Number of 1s = 26, so the prefix length is /26.

Host bits = 32 − 26 = 6

Block size = 256 − 192 = 64 (subnets in the last octet start at 0, 64, 128, 192)

Step 2: Network address of System 1 (IP address AND subnet mask)

IP address	11000000.10101000.00000001.10000010	(192.168.1.130)
Subnet mask	11111111.11111111.11111111.11000000	
AND result	11000000.10101000.00000001.10000000	= **192.168.1.128**

Step 3: Network address of System 2 (IP address AND subnet mask)

IP address	11000000.10101000.00000001.10111110	(192.168.1.190)
Subnet mask	11111111.11111111.11111111.11000000	
AND result	11000000.10101000.00000001.10000000	= **192.168.1.128**

Both systems give the same network address, so both belong to the same subnet 192.168.1.128/26.

Step 4: Broadcast address, subnet range and valid host range

Broadcast address = 128 + 64 − 1 = 191 → 192.168.1.191

Subnet range = 192.168.1.128 to 192.168.1.191

Valid host range = 192.168.1.129 to 192.168.1.190

Step 5: Does each IP address fall within the subnet range?

System 1: $129 \leq 130 \leq 190$ → Yes, 192.168.1.130 is a valid host address.

System 2: $129 \leq 190 \leq 190$ → Yes, 192.168.1.190 is a valid host address (the last usable address).

Step 6: Usable host addresses

Usable hosts = $2^6 - 2 = 64 - 2 = 62$ (in each subnet)

Final Answer:

Network address = 192.168.1.128 | Broadcast address = 192.168.1.191

Valid host range = 192.168.1.129 – 192.168.1.190 | Usable hosts = 62

Conclusion: Both IP addresses lie inside the same /26 subnet, so the two systems can communicate directly without a router.

Q3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

Solution:

Given: Network block = 172.16.0.0/16; HR = /25, Sales = /26, IT = /27, Admin = /28.

To find: Network address, host range, broadcast address and usable hosts of each department, and the unused range.

Step 1: Subnet masks, block size and usable hosts

HR (/25): 255.255.255.128; block = $2^7 = 128$; usable = $128 - 2 = 126$

Sales (/26): 255.255.255.192; block = $2^6 = 64$; usable = $64 - 2 = 62$

IT (/27): 255.255.255.224; block = $2^5 = 32$; usable = $32 - 2 = 30$

Admin (/28): 255.255.255.240; block = $2^4 = 16$; usable = $16 - 2 = 14$

Step 2: Allocation (in order, starting from 172.16.0.0)

HR: starts at 172.16.0.0

Network address = 172.16.0.0 /25

Broadcast address = $0 + 128 - 1 = 127 \rightarrow$ 172.16.0.127

Valid host range = 172.16.0.1 to 172.16.0.126

Sales: starts at $0 + 128 = 128$

Network address = 172.16.0.128 /26

Broadcast address = $128 + 64 - 1 = 191 \rightarrow$ 172.16.0.191

Valid host range = 172.16.0.129 to 172.16.0.190

IT: starts at $128 + 64 = 192$

Network address = 172.16.0.192 /27

Broadcast address = $192 + 32 - 1 = 223 \rightarrow$ 172.16.0.223

Valid host range = 172.16.0.193 to 172.16.0.222

Admin: starts at $192 + 32 = 224$

Network address = 172.16.0.224 /28

Broadcast address = $224 + 16 - 1 = 239 \rightarrow$ 172.16.0.239

Valid host range = 172.16.0.225 to 172.16.0.238

Step 3: Unused address range

Total addresses in a /16 block = $2^{16} = 65{,}536$

Addresses used = $128 + 64 + 32 + 16 = 240$ (172.16.0.0 to 172.16.0.239)

Unused addresses = $65{,}536 - 240 = 65{,}296$

The first free address is 172.16.0.240 and the block ends at 172.16.255.255.

Final Answer:

HR: 172.16.0.0/25 | .1 – .126 | Broadcast .127 | 126 hosts

Sales: 172.16.0.128/26 | .129 – .190 | Broadcast .191 | 62 hosts

IT: 172.16.0.192/27 | .193 – .222 | Broadcast .223 | 30 hosts

Admin: 172.16.0.224/28 | .225 – .238 | Broadcast .239 | 14 hosts

Unused range: 172.16.0.240 to 172.16.255.255 (65,296 addresses)

Conclusion: All four department subnets are continuous and non-overlapping, and most of the /16 block remains free for future departments.

Q4. A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

Solution:

Given: Network = 172.16.10.0/24; new subnet mask = /27 = 255.255.255.224.

To find: Number of subnets, usable hosts per subnet, subnet of 172.16.10.78 (network and broadcast address), and whether 172.16.10.95 is in the same subnet.

Step 1: Number of subnets

Subnet bits = 27 − 24 = 3

Number of subnets = $2^3 = 8$

Step 2: Usable hosts per subnet

Host bits = 32 − 27 = 5

Block size = $2^5 = 32$ (also 256 − 224 = 32)

Usable hosts = $2^5 - 2 = 32 - 2 = 30$

Step 3: The eight subnets (block size = 32)

Subnet 1: 172.16.10.0/27 (hosts .1 – .30, broadcast .31)
Subnet 2: 172.16.10.32/27 (hosts .33 – .62, broadcast .63)
Subnet 3: 172.16.10.64/27 (hosts .65 – .94, broadcast .95)
Subnet 4: 172.16.10.96/27 (hosts .97 – .126, broadcast .127)
Subnet 5: 172.16.10.128/27 (hosts .129 – .158, broadcast .159)
Subnet 6: 172.16.10.160/27 (hosts .161 – .190, broadcast .191)
Subnet 7: 172.16.10.192/27 (hosts .193 – .222, broadcast .223)
Subnet 8: 172.16.10.224/27 (hosts .225 – .254, broadcast .255)

Step 4: Subnet of 172.16.10.78 (IP address AND subnet mask)

IP address 10101100.00010000.00001010.01001110 (172.16.10.78)
Subnet mask 11111111.11111111.11111111.11100000
AND result 10101100.00010000.00001010.01000000 = **172.16.10.64**

Network address = 172.16.10.64

Broadcast address = 64 + 32 − 1 = 95 → 172.16.10.95

Valid host range = 172.16.10.65 to 172.16.10.94

Hence 172.16.10.78 belongs to Subnet 3 (172.16.10.64/27).

Step 5: Is 172.16.10.95 in the same subnet?

IP address 10101100.00010000.00001010.01011111 (172.16.10.95)
Subnet mask 11111111.11111111.11111111.11100000
AND result 10101100.00010000.00001010.01000000 = 172.16.10.64

The network address is the same, so 172.16.10.95 lies in the same subnet range (64 – 95). However, all its host bits are 1 (11111), so it is the broadcast address of the subnet and cannot be assigned to a host.

Final Answer:

Total subnets = 8 | Usable hosts per subnet = 30

172.16.10.78 → Subnet 172.16.10.64/27 (Broadcast 172.16.10.95)

172.16.10.95 → Same subnet range, but it is the broadcast address

Q5. A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Solution:

Given: IPv6 address = 2001:0db8:0000:0000:0000:ff00:0042:8329; prefix length = /64.

To find: Compressed form, expanded form, network prefix and number of host addresses.

Rules of compression:

(i) Leading zeros in each group may be removed.

(ii) One continuous run of all-zero groups may be replaced by "::" (only once in an address).

Step 1: Compression

Original address:

2001 : 0db8 : 0000 : 0000 : 0000 : ff00 : 0042 : 8329

Remove leading zeros in each group:

2001 : db8 : 0 : 0 : 0 : ff00 : 42 : 8329

Replace the three consecutive zero groups by "::":

2001:db8::ff00:42:8329

Step 2: Expansion of the compressed address

Groups present = 5 (2001, db8, ff00, 42, 8329); total groups required = 8.

Groups hidden by "::" = 8 − 5 = 3, so three groups of 0 are inserted:

2001:db8:0:0:0:ff00:42:8329

Pad every group to 4 hexadecimal digits:

2001:0db8:0000:0000:0000:ff00:0042:8329

This is the same as the given address, so the compression is verified.

Step 3: Network prefix for /64

64 bits = 4 groups of 16 bits, so the first four groups form the network prefix.

Network prefix = 2001:0db8:0000:0000 → **2001:db8::/64**

Interface identifier (remaining 64 bits) = 0000:ff00:0042:8329 → ::ff00:42:8329

Step 4: Number of host addresses

Host bits = 128 − 64 = 64

Number of addresses = $2^{64} = (2^{32})^2 = (4{,}294{,}967{,}296)^2$

= 18,446,744,073,709,551,616 (approximately 1.84×10^{19})

Final Answer:

Compressed form	**2001:db8::ff00:42:8329**
Expanded form	**2001:0db8:0000:0000:0000:ff00:0042:8329**
Network prefix	**2001:db8::/64**
Host addresses	**2^{64} = 18,446,744,073,709,551,616**

Conclusion: The address belongs to the network 2001:db8::/64. IPv6 has no broadcast address, so the count of 2^{64} addresses is not reduced by two as it is in IPv4.

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